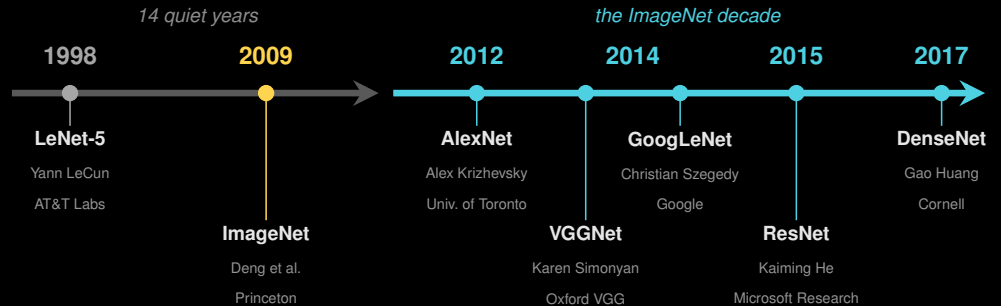
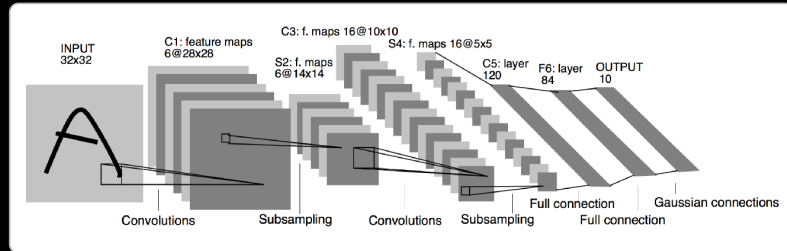


A short history of deep image classification

TWENTY YEARS, SIX ARCHITECTURES



LENET-5 (LECUN ET AL., 1998)



What made it the pioneering work

Automatic feature extraction

The convolutional layers **learned their own filters** to detect edges, corners and patterns.

Hierarchical representation

It processed data **in layers**, from simple visual patterns to complex digit shapes.

Why not an MLP

On raw pixels an MLP is intractable and wasteful, because pixels are **spatially correlated**.

Practical success

Remarkable accuracy on digit recognition, proving CNNs could **outperform traditional ML**.

LENET-5 RESULTS: MNIST



60,000 original images

Test error: **0.95%**

MNIST handwritten digits.



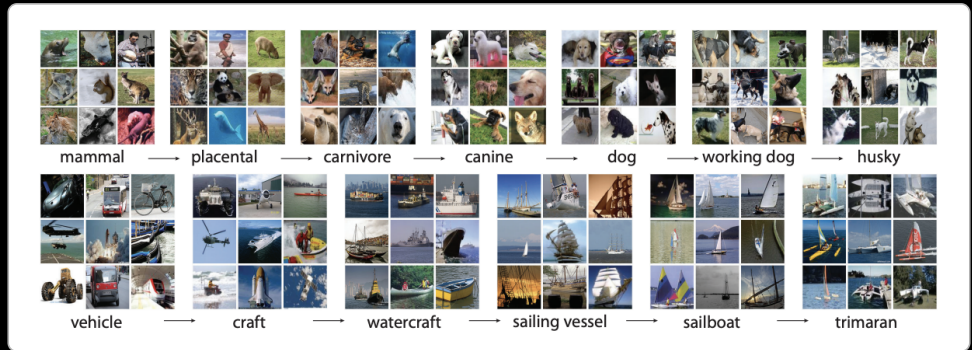
540,000 artificial distortions + 60,000 original

Test error: **0.8%**

The 82 misclassified test digits.

Distorting scarce data to get more of it worked already in 1998; Lecture 2 returns to this trick when medical labels get scarce.

IMAGENET FIXED THE DATA SIDE (2009)



Each row walks down the WordNet noun hierarchy, coarse to fine. ImageNet holds images at *every* node, not only at the leaves.

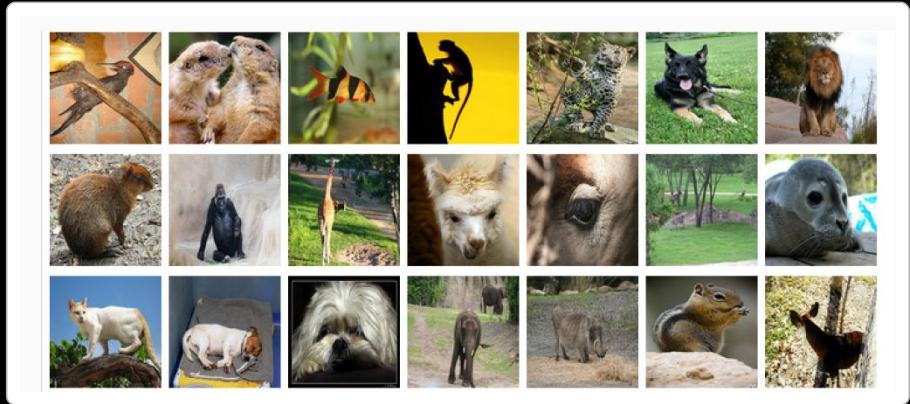
The scale, and the labels

- ▶ 1.2M images, 1000 classes, against 60k for CIFAR-10.
- ▶ Classes are **WordNet nouns**, crowdsourced rather than expert-annotated. Scale came from *cheap* labels.
- ▶ ILSVRC: an annual competition on a sealed test set, so every method was scored the same way.

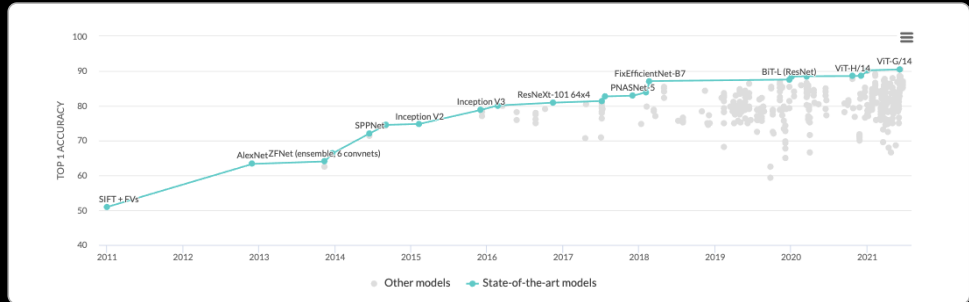
IMAGENET: THE CLASSIFICATION TASK

Classification goals

- ▶ Make 1 guess about the label: top-1 error.
- ▶ Make 5 guesses about the label: top-5 error.



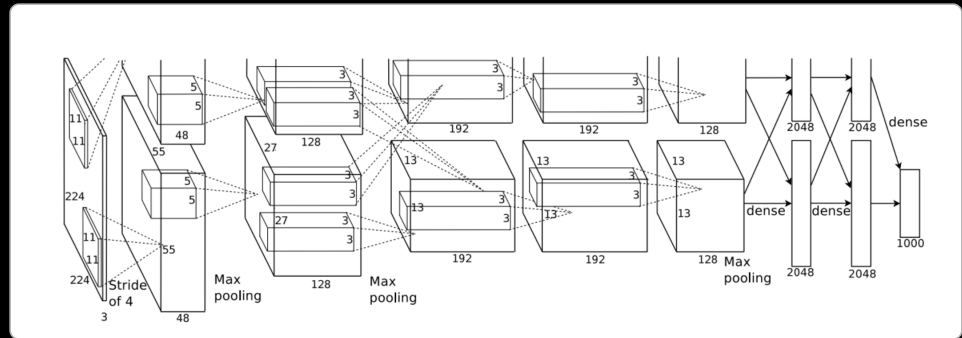
IMAGENET BECAME THE SCOREBOARD



Classical methods had hit a performance plateau. In 2012, Krizhevsky et al. took first place with their CNN-based network, **AlexNet**. Since then, the leaderboard has consisted of deep networks: first CNNs, and since 2020 vision transformers (the top points on this chart).

Axis note: this chart is top-1 *accuracy* (higher is better); the rest of this lecture quotes top-5 *error* (lower is better).

ALEXNET (2012): THE RESULT THAT RESTARTED THE FIELD



5 convolutional + 3 fully connected layers, split across two GPUs (the two rows).

At a glance

- ▶ **8 layers:** input $224 \times 224 \times 3$, 5 convolutional + 3 fully connected. About **60M** parameters.
- ▶ **Block:** conv $\rightarrow$ ReLU $\rightarrow$ max pool, with dropout in the FC stack. Split across two GPUs because it did not fit on one.
- ▶ **Result:** **16.4%** top-5 error against **26.2%** for the best non-CNN entry, a ten-point jump in a contest that usually moved by one or two.

WHAT ALEXNET ACTUALLY CHANGED

AlexNet keeps the same approach: successive filters designed to capture more and more subtle features. But the work explored several crucial details.

1. ReLU

Better nonlinearity: the derivative is 0 if the feature is below 0 and 1 for positive values. Efficient for gradient propagation.

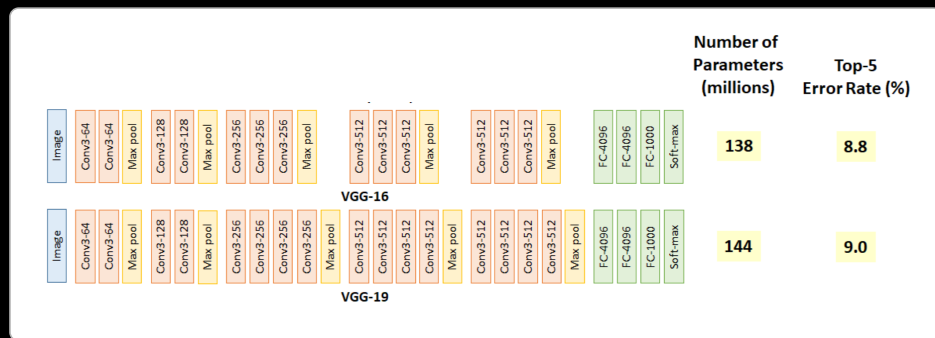
2. Dropout

Regularization: force the network to **forget things at random**, so that it sees the next input from a better perspective.

3. Augmentation

Images are shown with random **translation, rotation, crop**: the network learns the **attributes** of the images, rather than the images themselves.

VGGNET (2014): DEEP IS BETTER



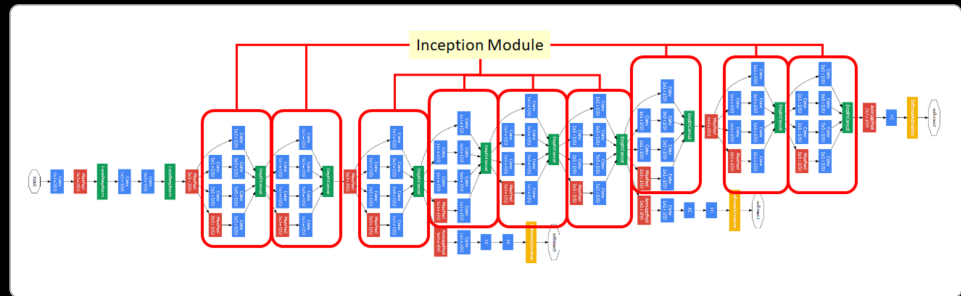
The two configurations everyone still uses. Full list in the paper.

Why small filters? Three reasons

1. Small filters induce **more nonlinearity**, which means more degrees of freedom for the network.
2. Stacking them lets the network **see more than it looks like**: two 3×3 layers see a 5×5 receptive field, three see 7×7 . The same feature extraction as with large filters.
3. Small filters also limit the number of parameters, which is good when you want to go that deep.

Quantitatively: 7.3% top-5 error on ImageNet.

GOOGLENET (2014): TIME FOR INCEPTION

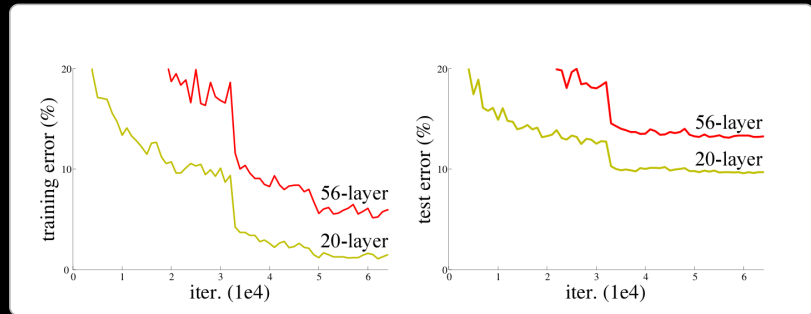


Nine stacked Inception modules (outlined in red), two auxiliary classifiers, and no large FC stack at the end.

Inception modules

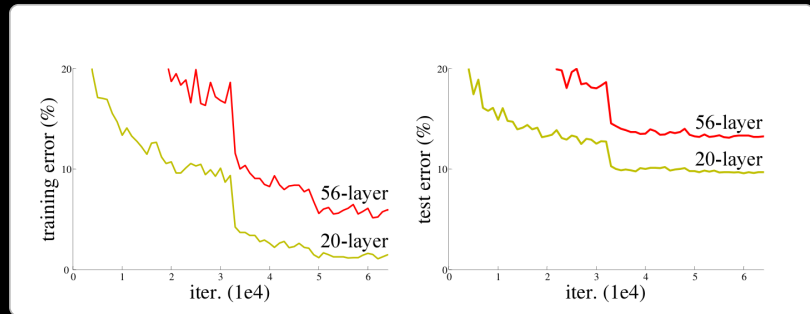
- ▶ Convolutions with **different filter sizes** are processed on the same input, then **concatenated**: multilevel feature extraction at each step. General features from the 5×5 filters, more local features from the 3×3 ones.
- ▶ Insanely expensive to compute? The brilliant solution: **1×1 convolutions**. They reduce the dimensionality of the features, and combine feature maps in a way that helps the representation.
- ▶ Why “inception”? The modules are **networks stacked inside a bigger network**. The best GoogLeNet ensemble: **6.7% top-5 error** on ImageNet.

AND THEN DEPTH STOPPED HELPING



Ask the room: what is wrong with these curves? Look at *both* panels before answering.

AND THEN DEPTH STOPPED HELPING



The degradation problem

With the network depth increasing, accuracy gets **saturated** (which might be unsurprising) and then **degrades rapidly**. Unexpectedly, such degradation is **not caused by overfitting**: adding more layers to a suitably deep model leads to higher training error. The degradation of training accuracy indicates that **not all systems are similarly easy to optimize**.

ResNet: making depth trainable

THE ARGUMENT THAT GIVES AWAY THE ANSWER

Hypothesis: the problem is an **optimization** problem; deeper models are harder to optimize.

In principle

The deeper model should be able to perform **at least as well** as the shallower model.

A solution by construction

Copy the **learned layers** from the shallower model, and set the additional layers to **identity mapping**.

learn $H(x)$ from scratch
plain layer

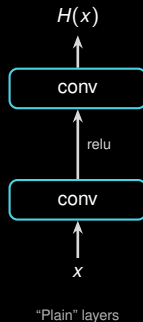


learn $\mathcal{F}(x) = H(x) - x$
residual layer

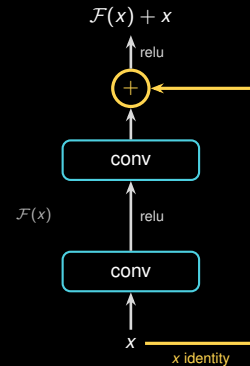
If the best thing to do is nothing,
the plain layer must *build* the identity;
the residual layer just sets $\mathcal{F} \rightarrow 0$.

THE RESIDUAL BLOCK

Structure each layer to fit a **residual function** with respect to the identity, then add the two back together.



"Plain" layers



Residual block

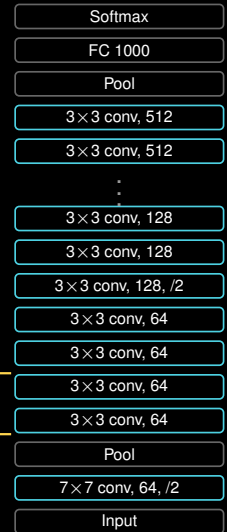
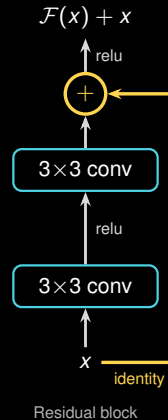
Residual learning: why this works

Every two layers, an identity mapping joins the output via **elementwise addition**. This is very helpful for **gradient propagation**: the error can be backpropagated through **multiple paths**. From a representation point of view, it also combines **different levels of features** at each step, just as the inception modules did. And it costs **no extra parameters**; it is an addition.

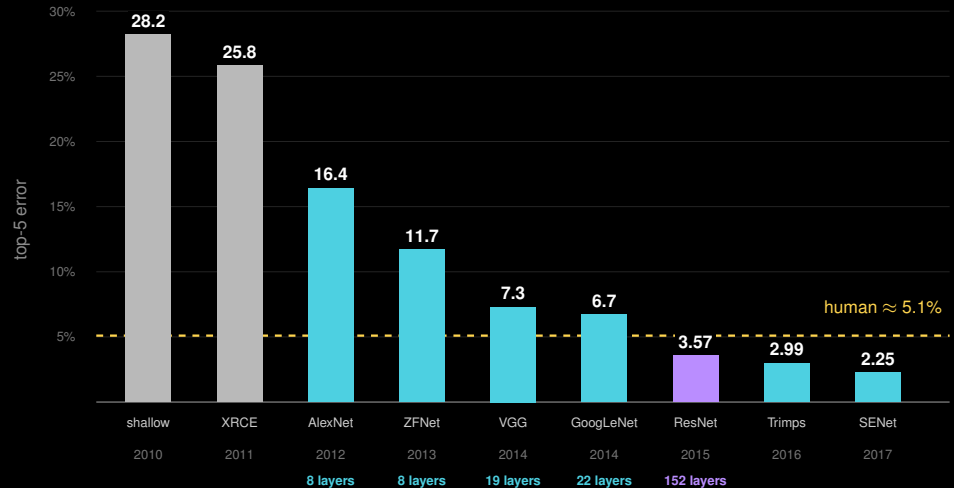
RESNET: A STACK OF RESIDUAL BLOCKS

Connect the layers

- ▶ All the networks so far followed the same trend: **going deeper**. But stacking more layers does not lead to better performance; the exact opposite occurs.
- ▶ Why? In one word: **the gradient**. The trick that counters it: **residual learning**.
- ▶ Every two layers, an **identity mapping** joins in via elementwise addition; the block on the left, stacked **18 to 152 layers deep**.
- ▶ To this date one of the best performing networks on ImageNet, with a **3.6% top-5 error rate**.

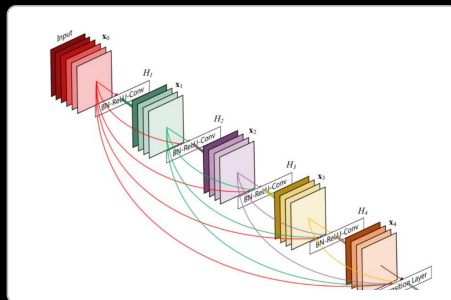


THE REVOLUTION OF DEPTH



Error fell 28% $\rightarrow$ 2% in seven years, and the depth column tells you how: 8 $\rightarrow$ 19 $\rightarrow$ 22 $\rightarrow$ 152 layers, once the residual connection made 152 trainable.

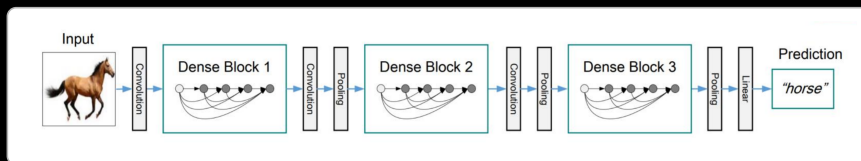
DENSENET (2017): CONNECT MORE!



Inside one dense block: every layer receives the feature maps of *all* preceding layers.

An extension of the reasoning

- ▶ DenseNet proposes **entire blocks of layers connected to one another**, which diversifies the features within those blocks far more.
- ▶ One character of difference:
ResNet: $x_\ell = \mathcal{F}(x_{\ell-1}) + x_{\ell-1}$ (add)
DenseNet: $x_\ell = \mathcal{F}([x_0, \dots, x_{\ell-1}])$ (concatenate)
- ▶ Nothing is ever summed away, so every layer sees the raw earlier features, and the net needs far fewer channels per layer.



CONCLUSION

Global trends

1. A major pattern observed overall: networks are designed to be **deeper and deeper**.
2. **Computational tricks** (ReLU, dropout, batch normalization) were introduced alongside them, with a significant impact on performance.
3. The apparition of **modules** able to capture **rich features** at each step of the network.
4. The increasing use of **connections between the layers**, which helps produce diverse features and proved useful for **gradient propagation**.

Next lecture: how to actually train one, and what to do when medical labels are scarce.